wheely v2 order

wheely v2: order checklist (2026-10-16)

Option B power (1S LiPo + LiPo Amigo Pro), breadboard first, soldered build after. Not needed again: Perma-Proto 3-pack, capacitor kit, 1N5819 diodes, M3 hardware, jumper wires, meter, soldering kit. Boards are not screwed down: the printed chassis holds the Perma-Proto in a tray with edge rails and the charger in a pocket; the Pico sits in its header socket. M3 only, for the mechanical parts.

The Pi Hut / Pimoroni

- ☐ 1 × Raspberry Pi Pico 2 WH (headers pre-soldered)
- ☐ 1 × Pimoroni LiPo Amigo Pro
- ☐ 1 × Pimoroni LiPo battery 2000 mAh, protected, JST-PH (1000 mAh if space is tight)
- ☐ 1 × Adafruit DRV8833 DC motor driver breakout
- ☐ 1 × RCWL-1601 ultrasonic sensor (3.3 V, HC-SR04 footprint)
- ☐ 1 × Full-size breadboard, 830 tie points
- ☐ 1 × 40-pin female header 2.54 mm (cut into 2 × 20 for the Pico socket)
- ☐ 2 × 40-pin male header strip 2.54 mm (driver breakout, Amigo Pro switch pads)

Amazon UK

- ☐ 1 × 2WD TT chassis kit (2 TT motors with wires, 2 wheels 65 mm, caster)
- ☐ 1 × 16 mm metal momentary push button with LED ring, 3 V LED version (12 V versions stay dark on a 3.7 V cell)
- ☐ 2 × 12 mm panel-mount momentary push button (reset; second one for the optional BOOTSEL)
- ☐ 1 × Panel-mount USB extension, USB-C female (panel) to micro-USB male, 20–30 cm
- ☐ 1 × JST-XH 2.54 mm connector kit: through-hole board headers + pre-crimped leads, 2/3/4-pin
- ☐ 1 × JST-PH 2.0 mm 2-pin pigtail pack (male + female)
- ☐ 1 × Female-female Dupont leads, 10 cm
- ☐ 1 × Silicone wire 22 AWG, red + black, 2 m each
- ☐ 1 × Silicone wire 26 AWG, five colours
- ☐ 1 × Heat-shrink tubing assortment
- ☐ 2 × 470 μF 10 V electrolytic capacitor (skip if the capacitor kit has 470 μF)
- ☐ 1 × Small zip ties

Decide later, after the breadboard tests

- [ ] 6 V step-up regulator rated 2 A or more (e.g. Pololu U3V40F6, check the rating): only if the motors are weak at cell voltage
- [ ] N20 6 V 150:1 gear motors with magnetic encoders × 2, matching 60 mm wheels and brackets (Pololu): the odometry upgrade for the map; changes the chassis
- [ ] 0.96” SSD1306 OLED, I²C: status display
- [ ] NeoPixel stick, 8 × WS2812: status light

Breadboard tests before soldering

1. Motors on carpet at cell voltage (3.3–4.2 V): decides the step-up.
2. Pico from the cell through the Schottky, USB plugged and unplugged: no reboot.
3. Power button through the Amigo Pro SW pads: one press on, one press off, LED follows.
4. Motor stall with both wheels held: if the Amigo Pro’s protection cuts out, feed the driver’s VM from the VBAT pin instead of the device output.
5. Reset via RUN; flash via `boot` over Bluetooth with the panel USB.